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November 11, 2025, 4:50 PM

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CAS Lexicon Search:

Appel reaction - Preferred Concept

AND Appel reaction - Preferred Concept

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Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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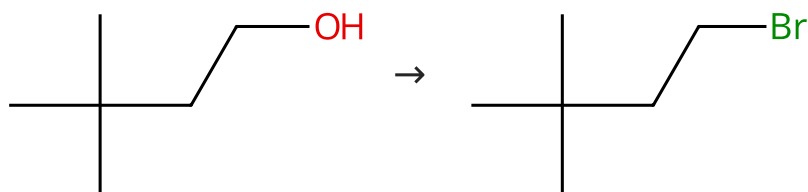
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (77)

 Suppliers (55)

31-614-CAS-31996428

Steps: 1 Yield: 100%

An environmentally benign and high-rate Appel type reaction

1.1 **Reagents:** 1,3-Dibromo-5,5-dimethylhydantoin, Triphenyl phosphine
Solvents: Dichloromethane; 0 °C; 15 min, 40 °C

By: Fuenzalida, Nicolas Mattias Del Rio; et al

Reaction Chemistry & Engineering (2022), 7(7), 1650-1659.

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



• HCl

• HCl

 Suppliers (28)

 Suppliers (85)

31-048-CAS-6037496

Steps: 1 Yield: 100%

Green chemical route for process development of lansoprazole intermediate

1.1 **Reagents:** Carbon tetrachloride, Triphenylphosphine
Solvents: Dichloromethane; 10 min, 10 - 15 °C

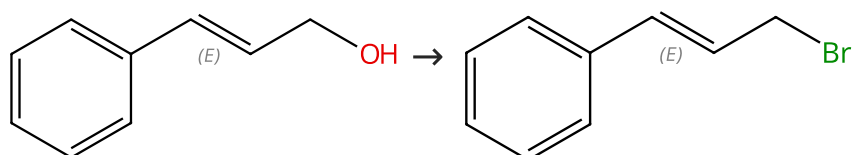
By: Pai, Nandini R.; et al

1.2 **Reagents:** Triethylamine
Catalysts: Tetramethylammonium bromide; 15 min, 15 - 20 °C; 20 °C → rt; 4 - 5 h, rt → 45 °C

Chemica Sinica (2013), 4(2), 122-126, 5 pp..

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

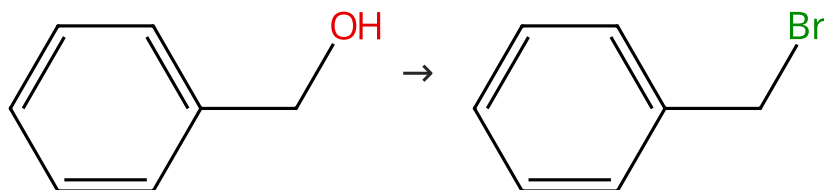
 Suppliers (66)

 Suppliers (38)

31-614-CAS-31996432	Steps: 1 Yield: 100%	An environmentally benign and high-rate Appel type reaction
1.1 Reagents: 1,3-Dibromo-5,5-dimethylhydantoin, Triphenyl phosphine Solvents: Dichloromethane; 0 °C; 15 min, 40 °C		By: Fuenzalida, Nicolas Mattias Del Rio; et al Reaction Chemistry & Engineering (2022), 7(7), 1650-1659.
Experimental Protocols		

Scheme 4 (2 Reactions)

Steps: 1 Yield: 97-100%



Suppliers (149)

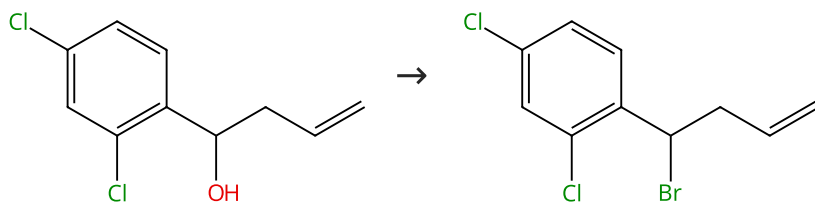
Suppliers (67)

31-614-CAS-31996436	Steps: 1 Yield: 100%	An environmentally benign and high-rate Appel type reaction
1.1 Reagents: 1,3-Dibromo-5,5-dimethylhydantoin, Triphenyl phosphine Solvents: Dichloromethane; 0 °C; 15 min, 40 °C		By: Fuenzalida, Nicolas Mattias Del Rio; et al Reaction Chemistry & Engineering (2022), 7(7), 1650-1659.
Experimental Protocols		

31-048-CAS-19760418	Steps: 1 Yield: 97%	1,2-Dibromotetrachloroethane: an efficient reagent for many transformations by modified Appel reaction
1.1 Reagents: Triphenylphosphine, 1,2-Dibromo-1,1,2,2-tetrachloroethane Solvents: Dichloromethane; 20 °C; 2 min, 20 °C		By: Essl, Selcuk; et al Turkish Journal of Chemistry (2019), 43(1), 150-156.
1.2 Solvents: Dichloromethane; 2 min, 20 °C; 5 min, 20 °C		

Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%

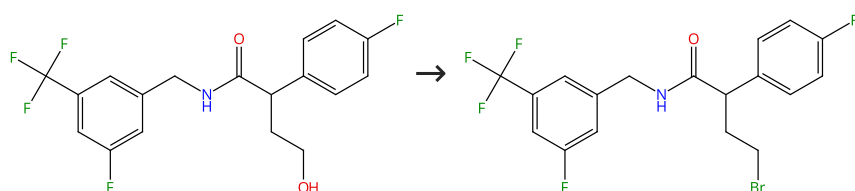


Suppliers (6)

31-048-CAS-13522552	Steps: 1 Yield: 99%	A mild and efficient method for bromination of alcohols using α,α -dibromo- β -dicarbonyl compounds as halogen sources
1.1 Reagents: Triphenylphosphine, Ethyl 2,2-dibromo-3-oxobutanoate Solvents: 1,2-Dichloroethane; 30 min, rt		By: Cui, Xiao-Meng; et al Tetrahedron Letters (2014), 55(1), 90-93.
1.2 Reagents: Water; rt		
Experimental Protocols		

Scheme 6 (1 Reaction)

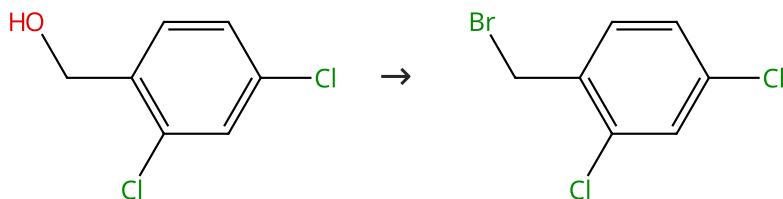
Steps: 1 Yield: 99%



31-048-CAS-11038413	Steps: 1 Yield: 99%	Synthesis and biological evaluation of spirocyclic antagonists of CCR2 (chemokine CC receptor subtype 2)
1.1 Reagents: Carbon tetrabromide, Triphenylphosphine Solvents: Acetonitrile; 30 min, 0 °C; overnight, rt		By: Strunz, Ann Kathrin; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2015), 23(14), 4034-4049.

Scheme 7 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (94)

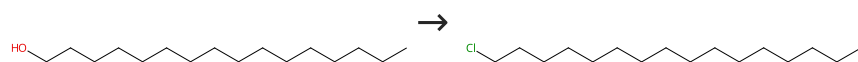
Suppliers (67)

31-048-CAS-5530222	Steps: 1 Yield: 99%	A mild and efficient method for bromination of alcohols using α,α-dibromo-β-dicarbonyl compounds as halogen sources
1.1 Reagents: Triphenylphosphine, 3,3-Dibromo-2,4-pentanedione Solvents: 1,2-Dichloroethane; 15 min, rt		By: Cui, Xiao-Meng; et al
1.2 Reagents: Water; rt		Tetrahedron Letters (2014), 55(1), 90-93.
Experimental Protocols		

31-048-CAS-1990510	Steps: 1 Yield: 99%	A mild and efficient method for bromination of alcohols using α,α-dibromo-β-dicarbonyl compounds as halogen sources
1.1 Reagents: Triphenylphosphine, Ethyl 2,2-dibromo-3-oxobutanoate Solvents: 1,2-Dichloroethane; 15 min, rt		By: Cui, Xiao-Meng; et al
1.2 Reagents: Water; rt		Tetrahedron Letters (2014), 55(1), 90-93.
Experimental Protocols		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



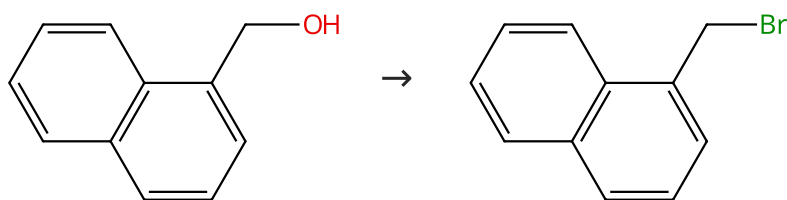
Suppliers (108)

Suppliers (56)

31-048-CAS-16147303	Steps: 1 Yield: 99%	Phosphine functionalized polyphosphazenes: soluble and reusable polymeric reagents for highly efficient halogenations under Appel conditions
1.1 Reagents: Triethylamine, 3-(Diphenylphosphino)-1-propanamine, Poly[nitrilo(dichlorophosphoranylidene)] Solvents: Tetrahydrofuran; 16 h, rt		By: Koenig, Michael; et al
1.2 Reagents: Hexachloroethane Solvents: Dichloromethane; 5 min, 0 °C; 15 min, 0 °C		Monatshefte fuer Chemie (2016), 147(9), 1575-1582.
Experimental Protocols		

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

Suppliers (65)

31-048-CAS-98761

Steps: 1 Yield: 99%

A mild and efficient method for bromination of alcohols using α,α -dibromo- β -dicarbonyl compounds as halogen sources

By: Cui, Xiao-Meng; et al

Tetrahedron Letters (2014), 55(1), 90-93.

1.1 **Reagents:** Triphenylphosphine, Ethyl 2,2-dibromo-3-oxobutanoate

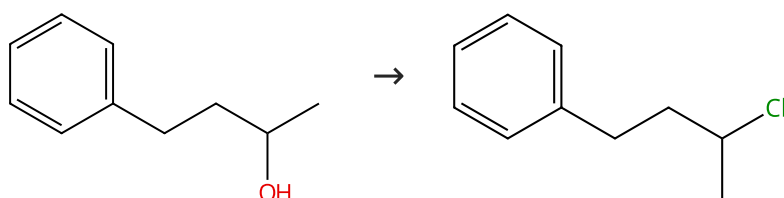
Solvents: 1,2-Dichloroethane; 30 min, rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (63)

Suppliers (7)

31-614-CAS-38945969

Steps: 1 Yield: 99%

Organocatalytic Stereospecific Appel Reaction

By: Toenjes, Jan; et al

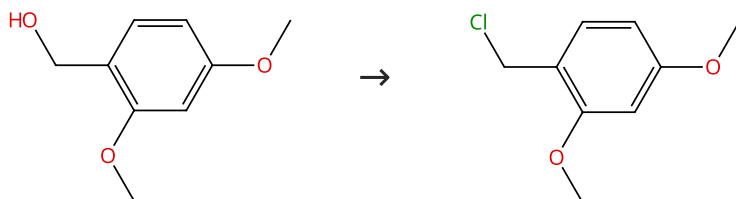
Organic Letters (2023), 25(51), 9114-9118.

1.1 **Reagents:** Hexachloroacetone, Phenylsilane
Catalysts: 3-Methyl-1-phenyl-2-phospholene 1-oxide; 24 h, 100 °C

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (32)

31-048-CAS-16147299

Steps: 1 Yield: 99%

Phosphine functionalized polyphosphazenes: soluble and reusable polymeric reagents for highly efficient halogenations under Appel conditions

By: Koenig, Michael; et al

Monatshefte fuer Chemie (2016), 147(9), 1575-1582.

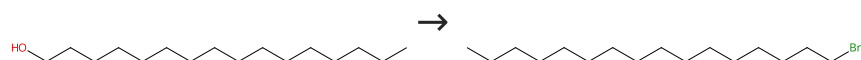
1.1 **Reagents:** Triethylamine, 3-(Diphenylphosphino)-1-propanamine, Poly[nitrilo(dichlorophosphoranylidene)]
Solvents: Tetrahydrofuran; 16 h, rt

1.2 **Reagents:** Hexachloroethane
Solvents: Dichloromethane; 5 min, 0 °C; 15 min, 0 °C

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (108)

Suppliers (75)

31-048-CAS-13194497

Steps: 1 Yield: 99%

A mild and efficient method for bromination of alcohols using α,α -dibromo- β -dicarbonyl compounds as halogen sources

By: Cui, Xiao-Meng; et al

Tetrahedron Letters (2014), 55(1), 90-93.

1.1 **Reagents:** Triphenylphosphine, Ethyl 2,2-dibromo-3-oxobutanoate

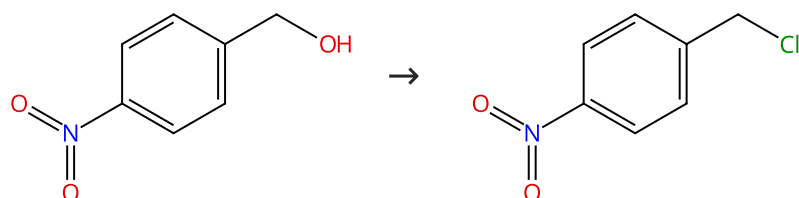
Solvents: 1,2-Dichloroethane; 15 min, rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (67)

31-048-CAS-16147300

Steps: 1 Yield: 99%

Phosphine functionalized polyphosphazenes: soluble and reusable polymeric reagents for highly efficient halogenations under Appel conditions

By: Koenig, Michael; et al

Monatshefte fuer Chemie (2016), 147(9), 1575-1582.

1.1 **Reagents:** Triethylamine, 3-(Diphenylphosphino)-1-propanamine, Poly[nitrilo(dichlorophosphoranylidene)]

Solvents: Tetrahydrofuran; 16 h, rt

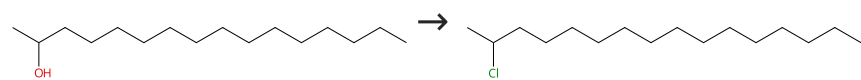
1.2 **Reagents:** Hexachloroethane

Solvents: Dichloromethane; 5 min, 0 °C; 15 min, 0 °C

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (46)

Suppliers (4)

31-048-CAS-16147304

Steps: 1 Yield: 98%

Phosphine functionalized polyphosphazenes: soluble and reusable polymeric reagents for highly efficient halogenations under Appel conditions

By: Koenig, Michael; et al

Monatshefte fuer Chemie (2016), 147(9), 1575-1582.

1.1 **Reagents:** Triethylamine, 3-(Diphenylphosphino)-1-propanamine, Poly[nitrilo(dichlorophosphoranylidene)]

Solvents: Tetrahydrofuran; 16 h, rt

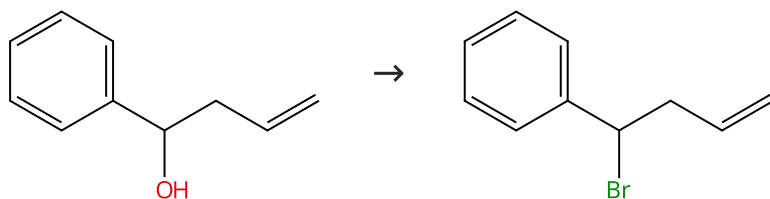
1.2 **Reagents:** Hexachloroethane

Solvents: Dichloromethane; 5 min, 0 °C; 15 min, 0 °C

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (69)

Suppliers (9)

31-048-CAS-15042844

Steps: 1 Yield: 98%

A mild and efficient method for bromination of alcohols using α,α -dibromo- β -dicarbonyl compounds as halogen sources

By: Cui, Xiao-Meng; et al

Tetrahedron Letters (2014), 55(1), 90-93.

1.1 **Reagents:** Triphenylphosphine, Ethyl 2,2-dibromo-3-oxobutanoate

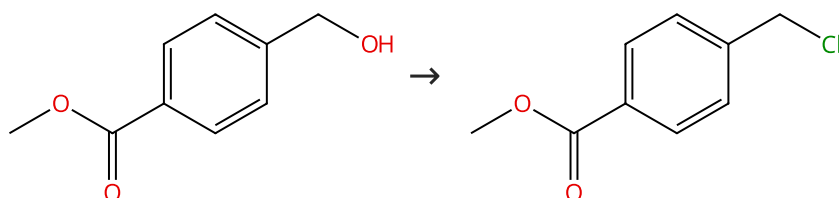
Solvents: 1,2-Dichloroethane; 30 min, rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (79)

Suppliers (71)

31-048-CAS-10500392

Steps: 1 Yield: 98%

The use of bromotrichloromethane in chlorination reactions

By: Newman, Stephen G.; et al

Synthesis (2011), (2), 342-346.

1.1 **Reagents:** Bromotrichloromethane, Triphenylphosphine

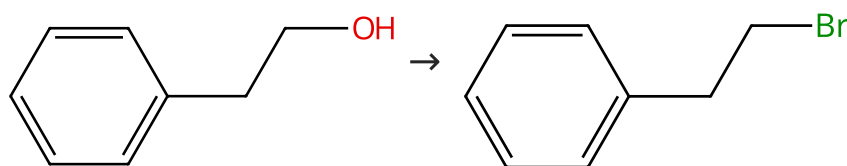
Solvents: Dichloromethane; 40 min, rt

1.2 **Solvents:** Dichloromethane; rt; 10 min, rt

Experimental Protocols

Scheme 17 (4 Reactions)

Steps: 1 Yield: 94-98%



Suppliers (110)

Suppliers (77)

31-614-CAS-31996429

Steps: 1 Yield: 98%

An environmentally benign and high-rate Appel type reaction

By: Fuenzalida, Nicolas Mattias Del Rio; et al

Reaction Chemistry & Engineering (2022), 7(7), 1650-1659.

1.1 **Reagents:** 1,3-Dibromo-5,5-dimethylhydantoin, Triphenyl phosphine

Solvents: Dichloromethane; > 0 °C; 30 min, 40 °C

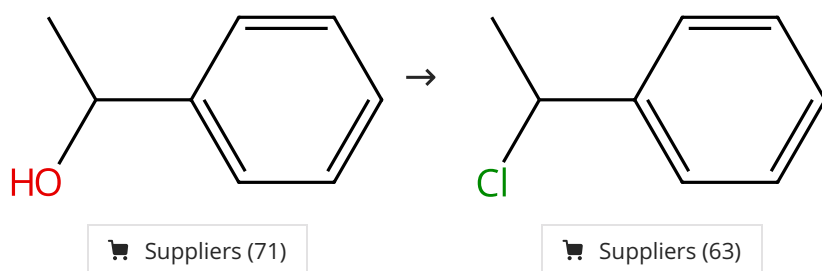
1.2 5 min, 40 °C

Experimental Protocols

31-048-CAS-19760419 Steps: 1 Yield: 97% 1.1 Reagents: Triphenylphosphine, 1,2-Dibromo-1,1,2,2-tetrachloroethane Solvents: Dichloromethane; 20 °C; 2 min, 20 °C 1.2 Solvents: Dichloromethane; 2 min, 20 °C; 5 min, 20 °C	1,2-Dibromotetrachloroethane: an efficient reagent for many transformations by modified Appel reaction By: Essl, Selcuk; et al Turkish Journal of Chemistry (2019), 43(1), 150-156.
31-614-CAS-44937118 Steps: 1 Yield: 95% 1.1 Reagents: Carbon tetrabromide, Phenylsilane Catalysts: <i>N,N'</i> -Bis(4-methoxyphenyl)- <i>P</i> -phenylphosphonic diamide Solvents: Acetonitrile; 24 h, 120 °C Experimental Protocols	Moderation of the Electronic Structure of Phosphamides to Execute the Catalytic Appel Reaction Bypassing Phosphine By: Kumari, Nidhi; et al Journal of Organic Chemistry (2024), 89(21), 15851-15863.
31-048-CAS-17626642 Steps: 1 Yield: 94% 1.1 Reagents: Carbon tetrabromide, Sodium bromide Catalysts: Tris(2,2'-bipyridine)ruthenium(2+) bis(hexafluoro phosphate) Solvents: Dimethylformamide; 16 h, rt Experimental Protocols	Approach to Comparing the Functional Group Tolerance of Reactions By: Gensch, Tobias; et al Journal of Organic Chemistry (2017), 82(17), 9154-9159.

Scheme 18 (1 Reaction)

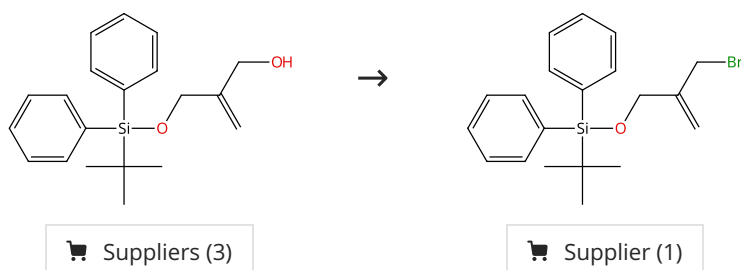
Steps: 1 Yield: 98%



31-048-CAS-6637387 Steps: 1 Yield: 98% 1.1 Reagents: Oxalyl chloride Catalysts: Triphenylphosphine oxide Solvents: Chloroform; 5 min, rt 1.2 Reagents: Oxalyl chloride Solvents: Chloroform; 7 h, rt Experimental Protocols	Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution Reactions: Development of a Catalytic Appel Reaction By: Denton, Ross M.; et al Journal of Organic Chemistry (2011), 76(16), 6749-6767.
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Scheme 19 (1 Reaction)

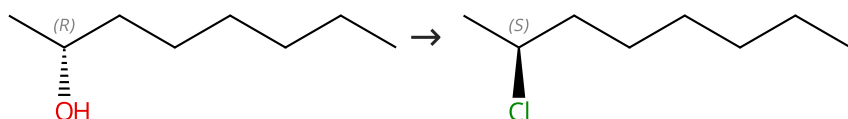
Steps: 1 Yield: 97%



31-048-CAS-16423659 Steps: 1 Yield: 97% 1.1 Reagents: Carbon tetrabromide, Triphenylphosphine Solvents: Dichloromethane; 1 h, 0 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water Experimental Protocols	Stereoselective synthesis of the (-)-dysiherbaine furopyran core via BF₃-promoted formal [3+2] annulation of aldehydo-aldoses with allylsilanes By: Hotoda, Koki; et al Tetrahedron Letters (2016), 57(48), 5359-5362.
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Scheme 20 (2 Reactions)

Steps: 1 Yield: 94-97%

Absolute stereochemistry shown,
Rotation (-)

Suppliers (79)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (5)

31-048-CAS-9103514

Steps: 1 Yield: 97%

Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution
Reactions: Development of a Catalytic Appel Reaction

By: Denton, Ross M.; et al

Journal of Organic Chemistry (2011), 76(16), 6749-6767.

- 1.1 Reagents: Oxalyl chloride
Catalysts: Triphenylphosphine oxide
Solvents: Chloroform; 5 min, rt
- 1.2 Reagents: Oxalyl chloride
Solvents: Chloroform; 7 h, rt

Experimental Protocols

31-048-CAS-6362376

Steps: 1 Yield: 94%

Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution
Reactions: Development of a Catalytic Appel Reaction

By: Denton, Ross M.; et al

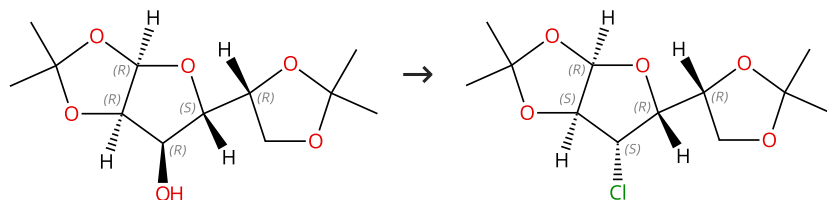
Journal of Organic Chemistry (2011), 76(16), 6749-6767.

- 1.1 Reagents: Oxalyl chloride
Catalysts: Tris(4-fluorophenyl)phosphine oxide
Solvents: Chloroform; 15 min, rt
- 1.2 Reagents: Oxalyl chloride
Solvents: Chloroform; 7 h, rt

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 96%

Absolute stereochemistry shown,
Rotation (+)

Suppliers (82)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (15)

31-048-CAS-1790705

Steps: 1 Yield: 96%

One pot conversion of carbohydrates alcohol into chloride via
benzotriazole sulfonate

By: Azad, Chandra S.; et al

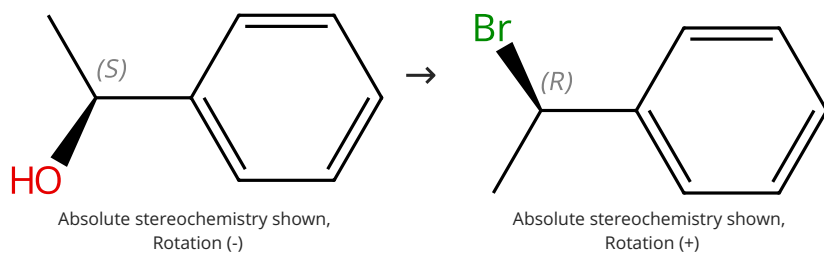
Tetrahedron (2013), 69(12), 2608-2612.

- 1.1 Reagents: Sulfuryl chloride
Solvents: Dichloromethane; -5 - 0 °C; 40 min, 0 °C
- 1.2 Reagents: 1*H*-Benzotriazole
Solvents: Dichloromethane; 0 °C; 6.5 h, rt

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (90)

Suppliers (23)

31-048-CAS-19760422

Steps: 1 Yield: 96%

1,2-Dibromotetrachloroethane: an efficient reagent for many transformations by modified Appel reaction

1.1 **Reagents:** Triphenylphosphine, 1,2-Dibromo-1,1,2,2-tetrachloroethane

Solvents: Dichloromethane; 20 °C; 2 min, 20 °C

1.2 **Solvents:** Dichloromethane; 2 min, 20 °C; 5 min, 20 °C

By: Esslitz, Selcuk; et al

Turkish Journal of Chemistry (2019), 43(1), 150-156.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (64)

Suppliers (57)

31-048-CAS-14757648

Steps: 1 Yield: 96%

The use of bromotrichloromethane in chlorination reactions

1.1 **Reagents:** Bromotrichloromethane, Triphenylphosphine

Solvents: Dichloromethane; 40 min, rt

1.2 **Solvents:** Dichloromethane; rt; 10 min, rt

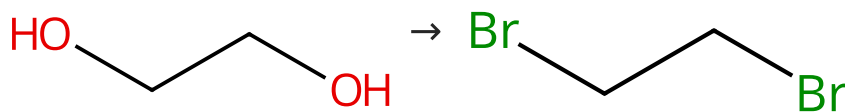
By: Newman, Stephen G.; et al

Synthesis (2011), (2), 342-346.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (166)

Suppliers (60)

31-048-CAS-19760421

Steps: 1 Yield: 96%

1,2-Dibromotetrachloroethane: an efficient reagent for many transformations by modified Appel reaction

1.1 **Reagents:** Triphenylphosphine, 1,2-Dibromo-1,1,2,2-tetrachloroethane

Solvents: Dichloromethane; 20 °C; 2 min, 20 °C

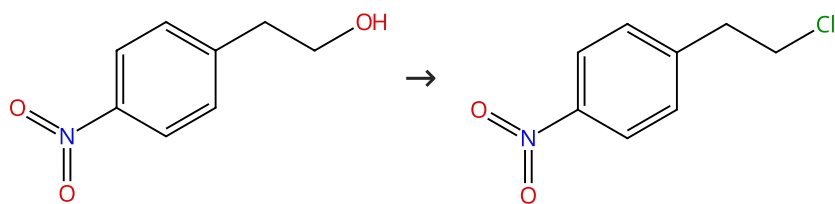
1.2 **Solvents:** Dichloromethane; 2 min, 20 °C; 5 min, 20 °C

By: Esslitz, Selcuk; et al

Turkish Journal of Chemistry (2019), 43(1), 150-156.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (85)

Suppliers (21)

31-614-CAS-38945972

Steps: 1 Yield: 96%

Organocatalytic Stereospecific Appel Reaction

- 1.1 **Reagents:** Hexachloroacetone, Phenylsilane
Catalysts: Phosphetane, 1,2,2,3,4,4-hexamethyl-, 1-oxide, *trans*-
Solvents: Toluene; 24 h, 100 °C

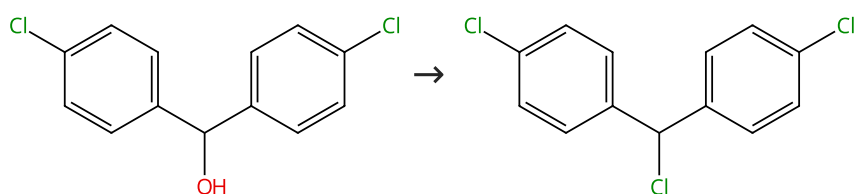
By: Toenjes, Jan; et al

Organic Letters (2023), 25(51), 9114-9118.

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (72)

Suppliers (60)

31-048-CAS-16147302

Steps: 1 Yield: 96%

Phosphine functionalized polyphosphazenes: soluble and re-usable polymeric reagents for highly efficient halogenations under Appel conditions

- 1.1 **Reagents:** Triethylamine, 3-(Diphenylphosphino)-1-propan amine, Poly[nitrilo(dichlorophosphoranylidene)]
Solvents: Tetrahydrofuran; 16 h, rt

By: Koenig, Michael; et al

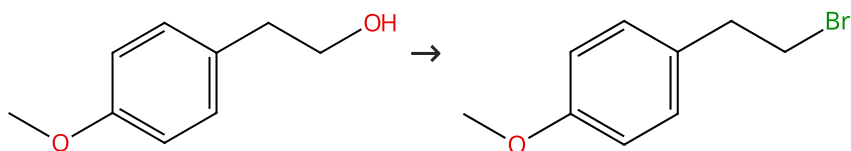
Monatshefte fuer Chemie (2016), 147(9), 1575-1582.

- 1.2 **Reagents:** Hexachloroethane
Solvents: Dichloromethane; 5 min, 0 °C; 15 min, 0 °C

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (85)

Suppliers (61)

31-614-CAS-44937127

Steps: 1 Yield: 96%

Moderation of the Electronic Structure of Phosphamides to Execute the Catalytic Appel Reaction Bypassing Phosphine

- 1.1 **Reagents:** Carbon tetrabromide, Phenylsilane
Catalysts: *N,N*-Bis(4-methoxyphenyl)-*P*-phenylphosphonic diamide
Solvents: Acetonitrile; 24 h, 120 °C

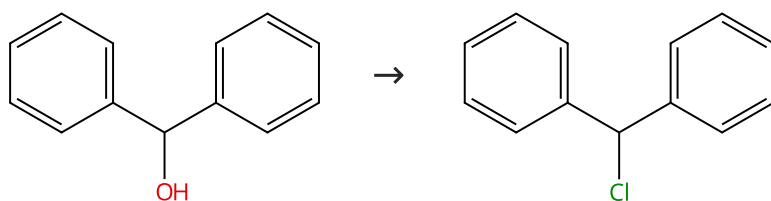
By: Kumari, Nidhi; et al

Journal of Organic Chemistry (2024), 89(21), 15851-15863.

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (96)

Suppliers (77)

31-048-CAS-8488980

Steps: 1 Yield: 96%

Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution Reactions: Development of a Catalytic Appel Reaction

By: Denton, Ross M.; et al

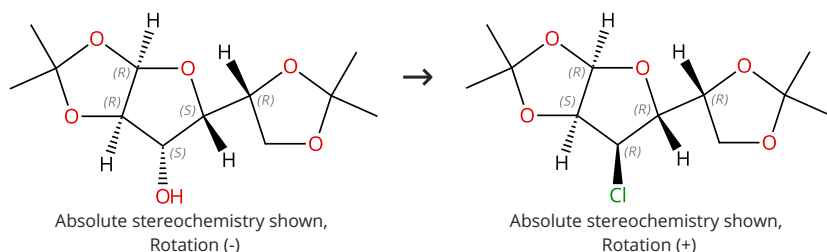
Journal of Organic Chemistry (2011), 76(16), 6749-6767.

- 1.1 **Reagents:** Oxalyl chloride
Catalysts: Triphenylphosphine oxide
Solvents: Chloroform; 5 min, rt
- 1.2 **Reagents:** Oxalyl chloride
Solvents: Chloroform; 7 h, rt

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (104)

31-048-CAS-8486199

Steps: 1 Yield: 95%

One pot conversion of carbohydrates alcohol into chloride via benzotriazole sulfonate

By: Azad, Chandra S.; et al

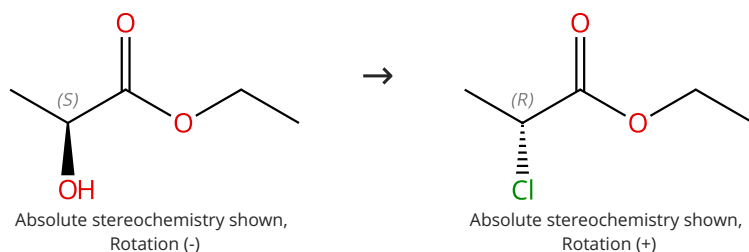
Tetrahedron (2013), 69(12), 2608-2612.

- 1.1 **Reagents:** Sulfuryl chloride
Solvents: Dichloromethane; -5 - 0 °C; 40 min, 0 °C
- 1.2 **Reagents:** 1*H*-Benzotriazole
Solvents: Dichloromethane; 0 °C; 6.5 h, rt

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 95%



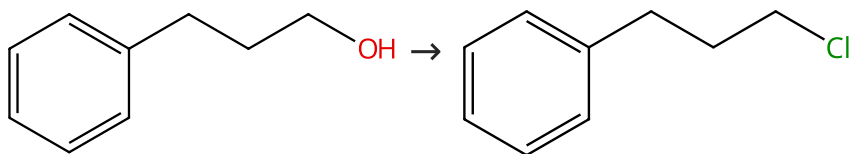
Suppliers (71)

Suppliers (16)

31-048-CAS-11229876	Steps: 1 Yield: 95%	Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution Reactions: Development of a Catalytic Appel Reaction By: Denton, Ross M.; et al Journal of Organic Chemistry (2011), 76(16), 6749-6767.
1.1 Reagents: Oxalyl chloride Catalysts: Triphenylphosphine oxide Solvents: Chloroform; 5 min, rt		
1.2 Reagents: Oxalyl chloride Solvents: Chloroform; 7 h, rt		
Experimental Protocols		

Scheme 31 (1 Reaction)

Steps: 1 Yield: 95%



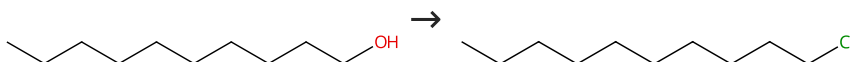
Suppliers (86)

Suppliers (66)

31-048-CAS-13360873	Steps: 1 Yield: 95%	Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution Reactions: Development of a Catalytic Appel Reaction By: Denton, Ross M.; et al Journal of Organic Chemistry (2011), 76(16), 6749-6767.
1.1 Reagents: Oxalyl chloride Catalysts: Triphenylphosphine oxide Solvents: Chloroform; 5 min, rt		
1.2 Reagents: Oxalyl chloride Solvents: Chloroform; 7 h, rt		
Experimental Protocols		

Scheme 32 (3 Reactions)

Steps: 1 Yield: 94-95%



Suppliers (91)

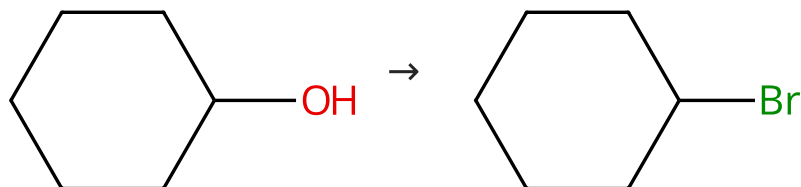
Suppliers (68)

31-048-CAS-167215	Steps: 1 Yield: 95%	A procedure for Appel halogenations and dehydrations using a polystyrene supported phosphine oxide By: Tang, Xiaoping; et al Tetrahedron Letters (2014), 55(4), 799-802.
1.1 Reagents: Phosphorus(1+), chlorotriphenyl-, chloride (1:1), (7-4)- (polystyrene-supported) Solvents: Chloroform; 18 h, 60 °C		
Experimental Protocols		

31-048-CAS-9100673	Steps: 1 Yield: 95%	Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution Reactions: Development of a Catalytic Appel Reaction By: Denton, Ross M.; et al Journal of Organic Chemistry (2011), 76(16), 6749-6767.
1.1 Reagents: Oxalyl chloride Catalysts: Tris(4-fluorophenyl)phosphine oxide Solvents: Chloroform; 15 min, rt		
1.2 Reagents: Oxalyl chloride Solvents: Chloroform; 7 h, rt		
Experimental Protocols		

31-048-CAS-4837915	Steps: 1 Yield: 94%	Catalytic Phosphorus(V)-Mediated Nucleophilic Substitution Reactions: Development of a Catalytic Appel Reaction
1.1 Reagents: Oxalyl chloride Catalysts: Triphenylphosphine oxide Solvents: Chloroform; 5 min, rt		By: Denton, Ross M.; et al
1.2 Reagents: Oxalyl chloride Solvents: Chloroform; 7 h, rt		Journal of Organic Chemistry (2011), 76(16), 6749-6767.
Experimental Protocols		

Scheme 33 (1 Reaction)

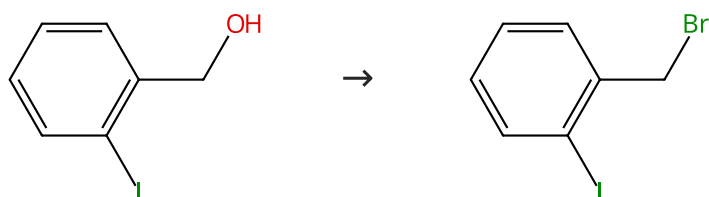
Steps: **1** Yield: **95%**

Suppliers (81)

Suppliers (73)

31-048-CAS-19760423	Steps: 1 Yield: 95%	1,2-Dibromotetrachloroethane: an efficient reagent for many transformations by modified Appel reaction
1.1 Reagents: Triphenylphosphine, 1,2-Dibromo-1,1,2,2-tetrachloroethane Solvents: Dichloromethane; 20 °C; 2 min, 20 °C		By: Esslitz, Selcuk; et al
1.2 Solvents: Dichloromethane; 2 min, 20 °C; 5 min, 20 °C		Turkish Journal of Chemistry (2019), 43(1), 150-156.

Scheme 34 (1 Reaction)

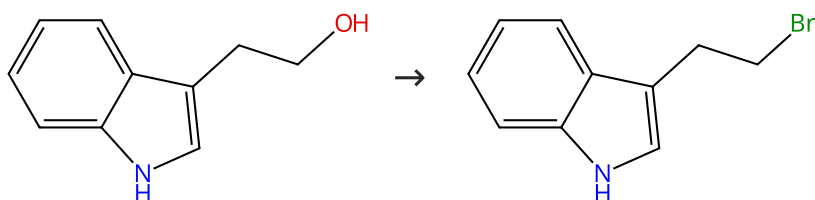
Steps: **1** Yield: **95%**

Suppliers (87)

Suppliers (67)

31-048-CAS-2223749	Steps: 1 Yield: 95%	The application of a monolithic triphenylphosphine reagent for conducting Appel reactions in flow microreactors
1.1 Reagents: Triphenylphosphine Solvents: Dichloromethane; 1.25 h, rt		By: Roper, Kimberley A.; et al
		Beilstein Journal of Organic Chemistry (2011), 7, 1648-1655.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **95%**

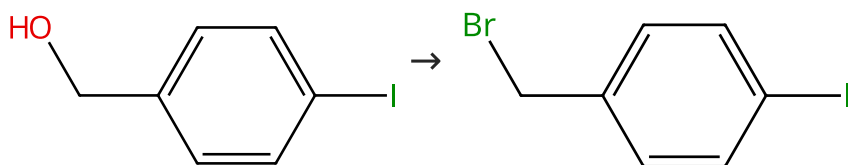
Suppliers (94)

Suppliers (72)

31-048-CAS-12301154 Steps: 1 Yield: 95% 1.1 Reagents: Phosphonium, (dibromomethyl)triphenyl-, inner salt (polymer-supported phosphonium dibromomethylide, phosphonium ylide) Solvents: Dichloromethane; 75 min, 100 psi, 0 °C	The application of a monolithic triphenylphosphine reagent for conducting Ramirez gem-dibromoolefination reactions in flow By: Roper, Kimberley A.; et al Beilstein Journal of Organic Chemistry (2013), 9, 1781-1790, 10 pp..
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Scheme 36 (1 Reaction)

Steps: 1 Yield: 95%



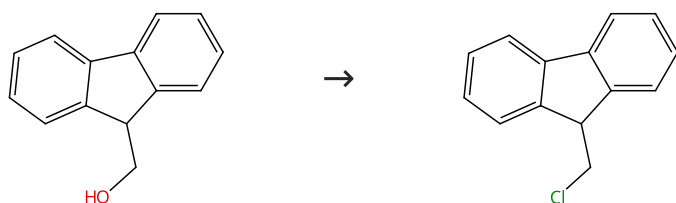
Suppliers (85)

Suppliers (84)

31-048-CAS-13478281 Steps: 1 Yield: 95% 1.1 Reagents: Triphenylphosphine Solvents: Dichloromethane; 1.25 h, rt	The application of a monolithic triphenylphosphine reagent for conducting Appel reactions in flow microreactors By: Roper, Kimberley A.; et al Beilstein Journal of Organic Chemistry (2011), 7, 1648-1655.
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Scheme 37 (1 Reaction)

Steps: 1 Yield: 95%



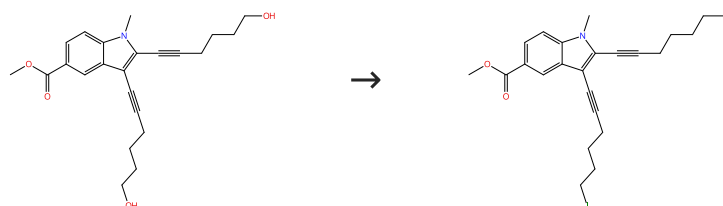
Suppliers (88)

Suppliers (18)

31-614-CAS-38945975 Steps: 1 Yield: 95% 1.1 Reagents: Hexachloroacetone, Phenylsilane Catalysts: Phosphetane, 1,2,2,3,4,4-hexamethyl-, 1-oxide, <i>trans</i>- Solvents: Toluene; 24 h, 100 °C Experimental Protocols	Organocatalytic Stereospecific Appel Reaction By: Toenjes, Jan; et al Organic Letters (2023), 25(51), 9114-9118.
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Scheme 38 (1 Reaction)

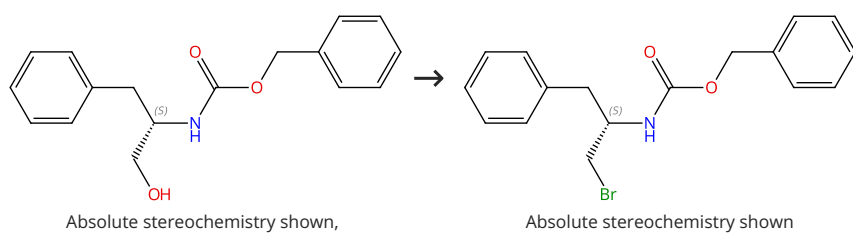
Steps: 1 Yield: 94%



31-048-CAS-9541532 Steps: 1 Yield: 94% 1.1 Reagents: Imidazole, Triphenylphosphine, Iodine Solvents: Tetrahydrofuran; 1 h, -20 °C; -20 °C → rt Experimental Protocols	Electrophilic Cyclization of Aryldiacetylenes in the Synthesis of Functionalized Enediynes Fused to a Heterocyclic Core By: Danilkina, N. A.; et al Journal of Organic Chemistry (2014), 79(19), 9018-9045.
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Scheme 39 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (72)

Suppliers (2)

31-614-CAS-24161496

Steps: 1 Yield: 94%

N-Bromosuccinimide promoted synthesis of β -amino bromides under Appel reaction condition
1.1 Reagents: *N*-Bromosuccinimide, Triphenylphosphine

Solvents: Dichloromethane; 15 min, 0 °C; 2.5 h, rt

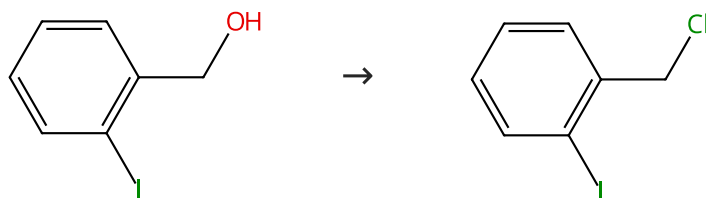
By: Chinthaginjala, Srinivasulu; et al

Experimental Protocols

Synthetic Communications (2021), 51(19), 2975-2983.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (87)

Suppliers (58)

31-048-CAS-8372484

Steps: 1 Yield: 94%

The use of bromotrichloromethane in chlorination reactions

1.1 Reagents: Bromotrichloromethane, Triphenylphosphine

Solvents: Dichloromethane; 40 min, rt

By: Newman, Stephen G.; et al

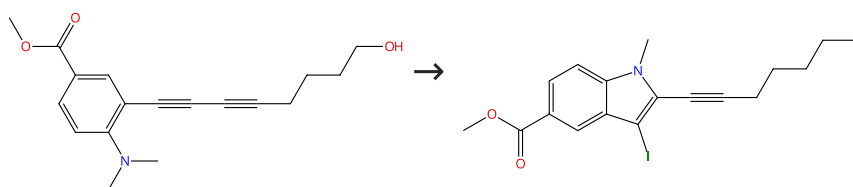
1.2 Solvents: Dichloromethane; rt; 10 min, rt

Synthesis (2011), (2), 342-346.

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 94%



31-048-CAS-15111772

Steps: 1 Yield: 94%

Electrophilic Cyclization of Aryldiacetylenes in the Synthesis of Functionalized Enediynes Fused to a Heterocyclic Core

1.1 Reagents: Imidazole, Triphenylphosphine, Iodine

Solvents: Dichloromethane; 1 h, -20 °C; -20 °C → rt

By: Danilkina, N. A.; et al

1.2 Reagents: Iodine

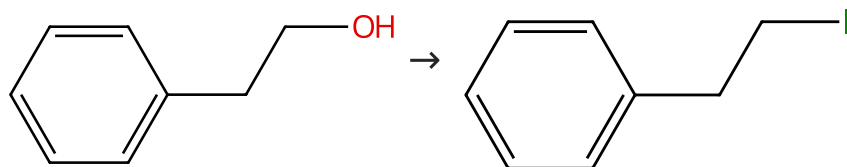
Solvents: Dichloromethane; rt; 18 h, reflux

Journal of Organic Chemistry (2014), 79(19), 9018-9045.

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (110)

Suppliers (58)

31-614-CAS-31996435

Steps: 1 Yield: 94%

An environmentally benign and high-rate Appel type reaction1.1 **Reagents:** *N*-Iodosuccinimide, Triphenylphosphine

By: Fuenzalida, Nicolas Mattias Del Rio; et al

Solvents: Dichloromethane; 0 °C; 5 h, 40 °C

Reaction Chemistry & Engineering (2022), 7(7), 1650-1659.

Experimental Protocols